

Gagan Bansal

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Summary

Computer Science Graduate with R&D internship at Quark Software. Hands-on experience in backend systems, applied AI development, and cross-platform C++ porting.

Technical Skills

Languages: C++, Go, Python

Web & Backend: React.js, TailwindCSS, Node.js, Express.js, FastAPI, RESTful APIs, Kafka

Databases: PostgreSQL, MongoDB, Redis, SQL

GenAI: LLM APIs, RAG Pipelines, Pinecone, LangChain, LangGraph, Agentic AI

Tools: Figma, Git, Linux, macOS, Claude Code

Experience

Research and Development Intern, Quark Software

Jan 2026 – June 2026 *Mohali*

- Ported and modernized a legacy codebase to ensure compatibility with the latest macOS versions, supporting both ARM64 (Apple Silicon) and x86-64 architectures.
- Resolved complex cross-platform build and dependency issues involving OpenSSL and Axis2, ensuring stable builds across ARM64 and x86 architectures.
- Debugged and fixed low-level compatibility issues arising from architecture differences and deprecated system APIs.

Projects

Genei AI

GitHub

Python, LangChain, Pinecone, Google Gemini API, React.Js, RAG, Agent Tools

- Developed Genei AI, an end-to-end AI knowledge platform that transforms documents into an intelligent knowledge base using RAG, enabling accurate, context-aware conversations with source citations.
- Architected a scalable and modular backend using FastAPI, LangChain, Pinecone, PostgreSQL, and Google Gemini, supporting document ingestion, semantic retrieval, persistent chat sessions, and clean, extensible system design.
- Supporting multi-document knowledge retrieval and extensibility for AI agents, tool calling, hybrid search, re-ranking, and response streaming.

KVMemo – Multithreaded In-Memory Key-Value Store

demo

C++

- Designed and implemented a high-performance in-memory key-value store in modern C++ supporting concurrent TCP client connections.
- Built a thread-pool architecture with task scheduling and synchronization using `std::mutex`, `std::shared-mutex`, and atomic counters to optimize read-heavy workloads.

oForge - Engineering Workspace Platform

demo

React.js, Node.js, Express.js, MongoDB, PostgreSQL, TailwindCSS

- Built oForge as a team collaboration platform for students, combining team discovery with a feature-rich workspace including Kanban boards, real-time chat, schema design, documentation, and whiteboard tools.
- Introduced a gated team collaboration system where users must submit proposals to join projects, with workspace access granted only after owner approval, ensuring privacy and controlled team formation.

Education

Chitkara University, Chandigarh

Oct 2022 – June 2026

B.E Computer Science

CGPA: 9.44

Achievements

- Solved **1300+** DSA problems across LeetCode, GFG, and CodeChef & **LeetCode Contest Rating: 1618**.
- Selected for UCA Batch at Chitkara University – Top 3% out of 2500+ students.
- Runner Up** – Shopclues Smart AI Cataloging Hackathon (Top 5 of 70+ teams), Chandigarh, 2024.